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Abhirup Sarker

Lecturer

Department of Robotics and Mechatronics

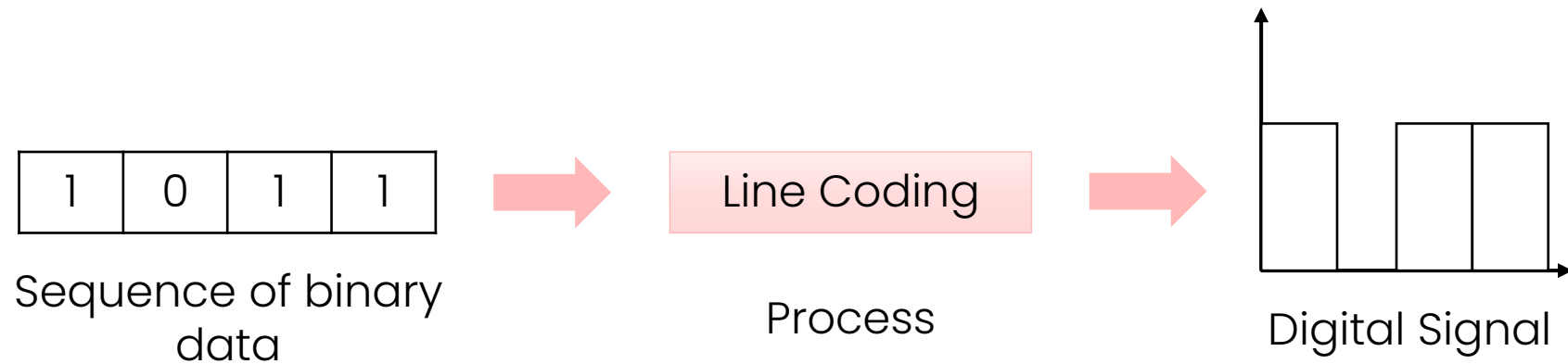
Engineering

Daffodils International University



LINE CODING

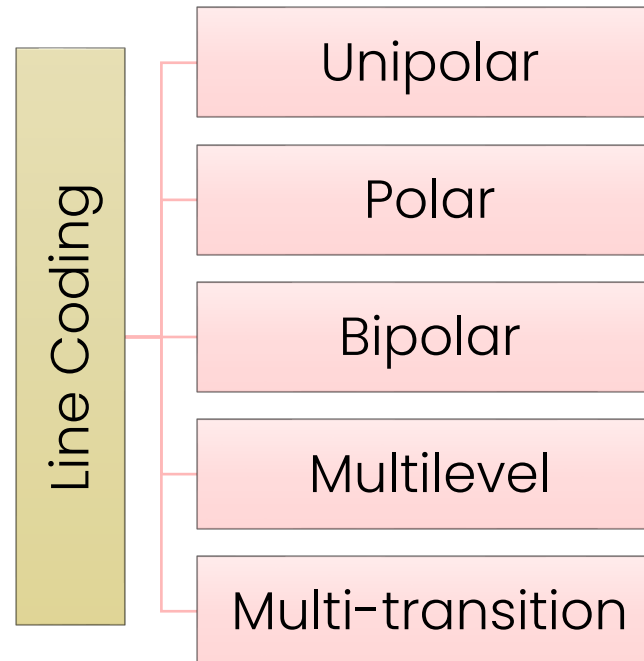
- ❑ **Line Coding** is the process of converting binary data into a sequence of bits of the digital signal.
- ❑ Everyday we come across various types like videos, text, images, numerical, signal which are stored as 1 & 0 in our computers. Line coding converts these binary sequences to digital signal.



LINE CODING

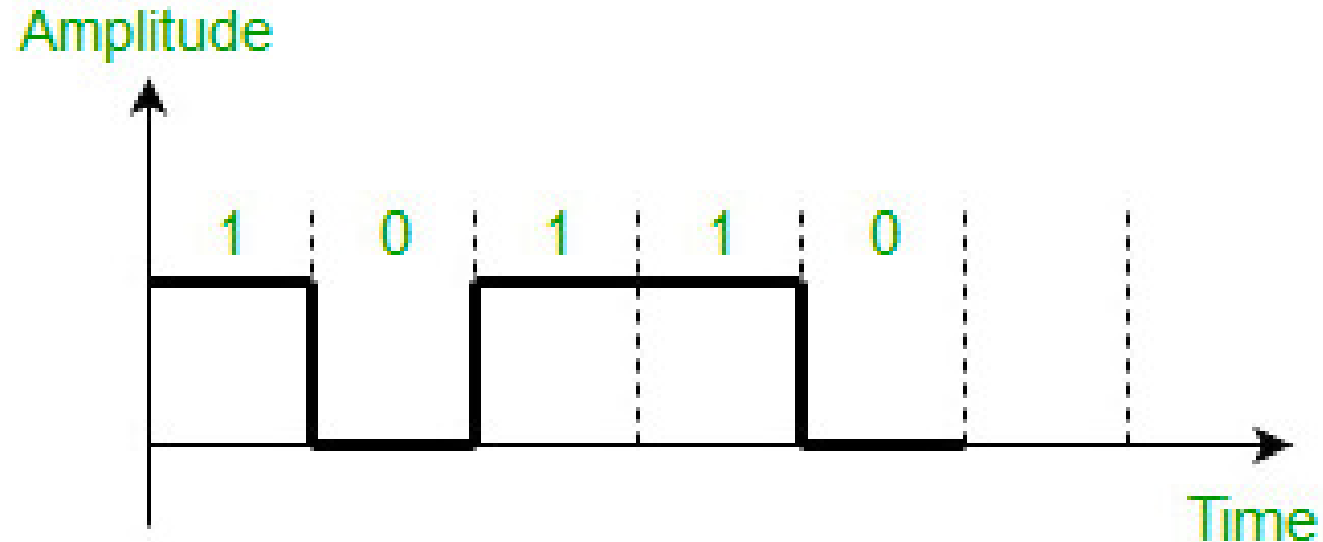
Characteristics:

1. Noise immunity
2. Long range transmission
3. Self-synchronization of data
4. Bandwidth compression
5. Minimum cross-talk
6. Easy error detection



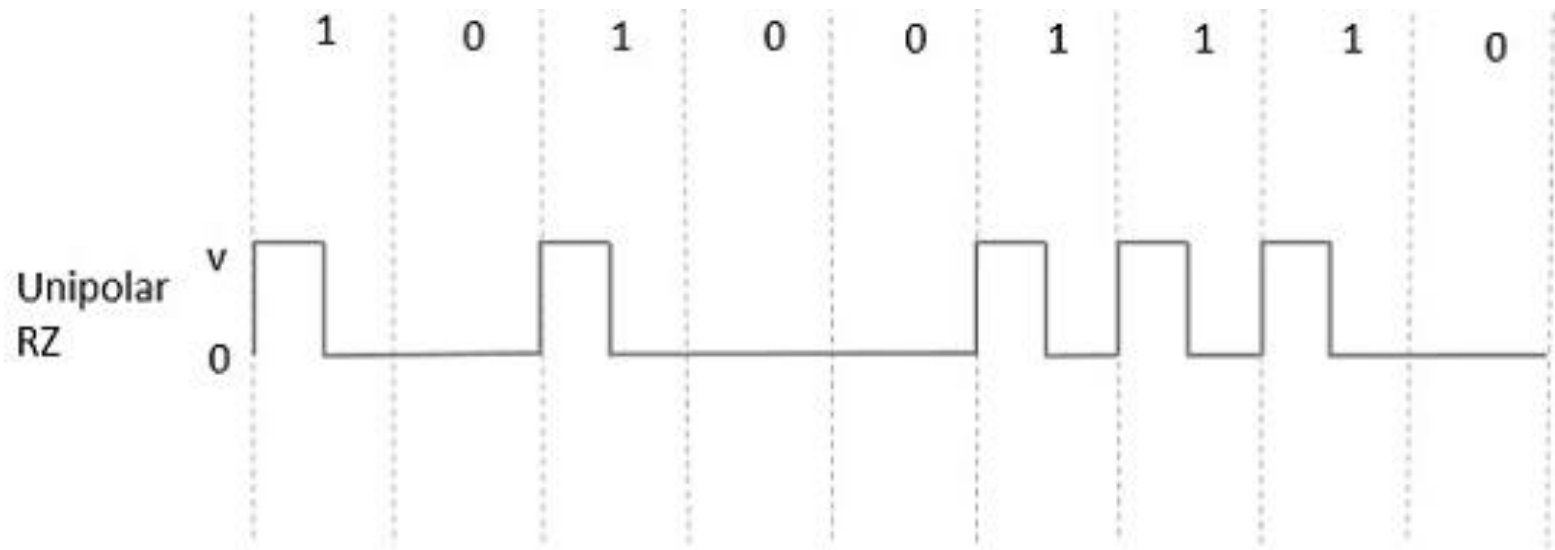
UNIPOLAR

- ❑ NRZ or Non-Return to Zero is a unipolar line coding scheme. Here data bit 0 presents voltage 0 and data bit 1 represent positive voltage. The signal does not return to zero at the middle of time T for the bit, hence it is called NRZ.



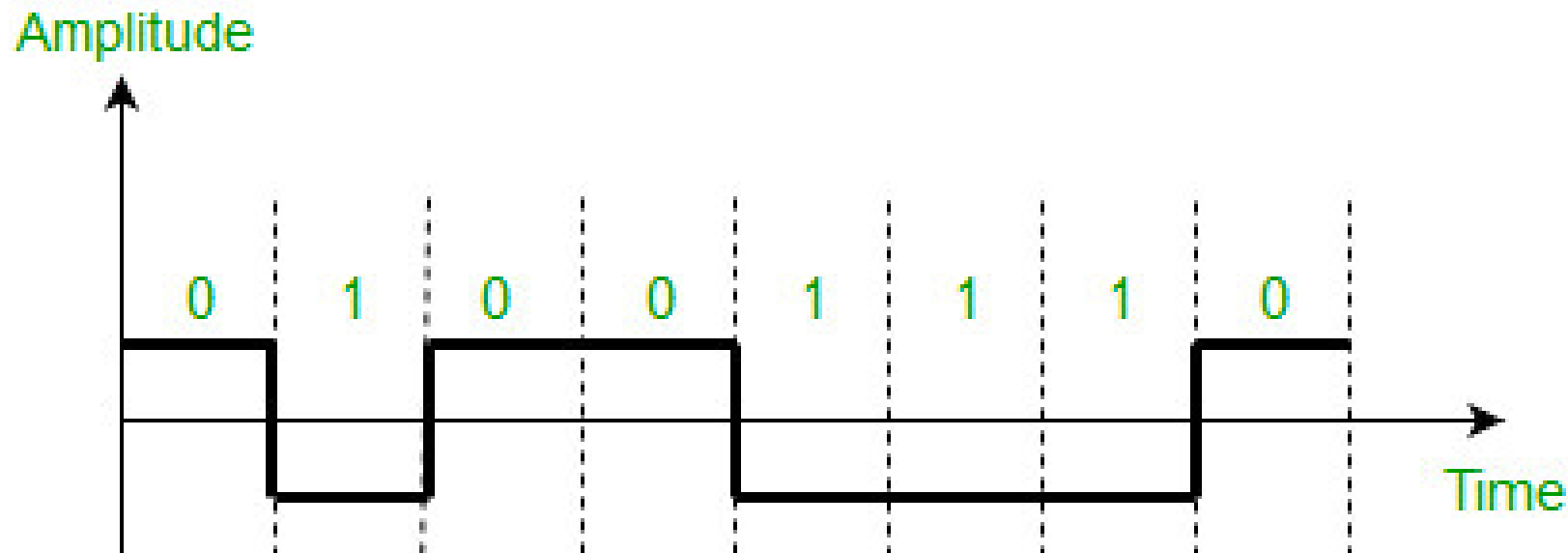
UNIPOLAR

- ❑ In RZ or Return to Zero the signal returns to zero at $T/2$ for bit 1. There is no change for bit 0. Here also bit 1 represents positive voltage and bit 0 represent zero voltage.



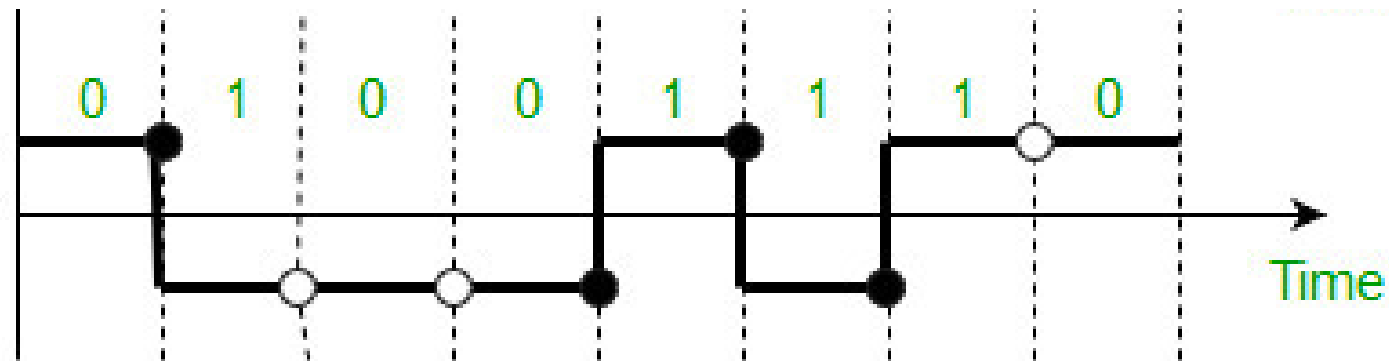
POLAR

- In polar line coding there are two types of NRZ: 1. NRZ-L(level) and 2. NRZ-I(invert). In case of NRZ-Level generally bit 0 represents logic high/ positive voltage and bit 1 represents logic low/ negative voltage.



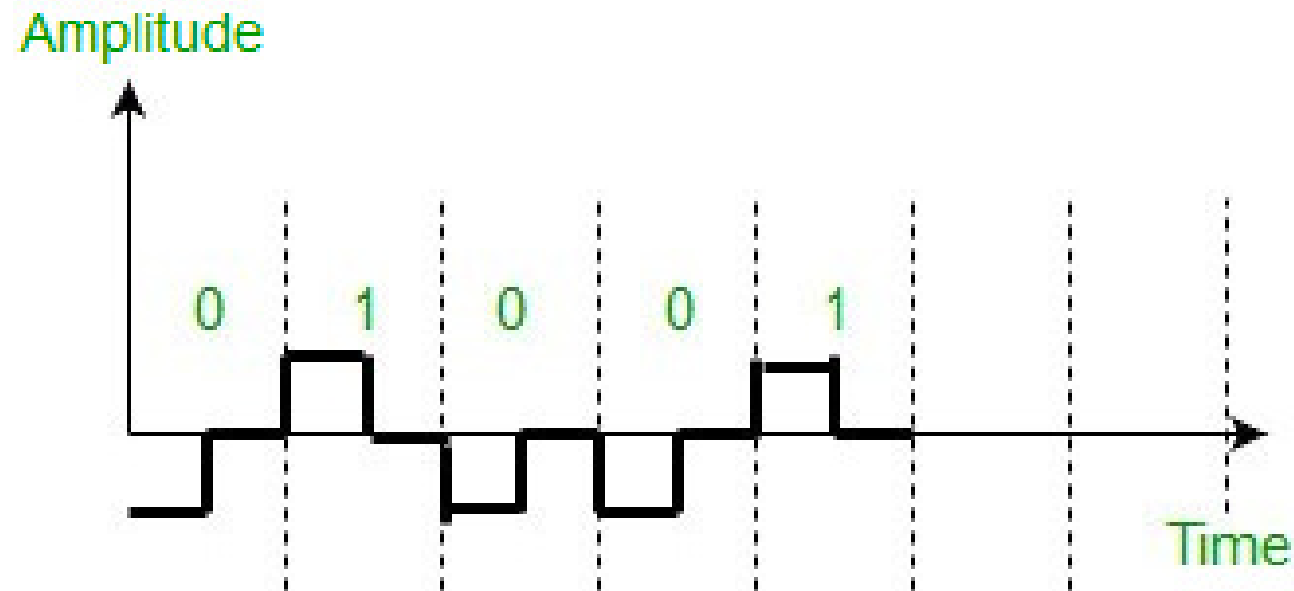
POLAR

- ❑ In **NRZ-I(invert)**, the signal alternates for bit 1 and for bit 0 the signal stream remains as it is. We indicate the transition by bold dots and unchanged states with hollow dots in the digital signal.



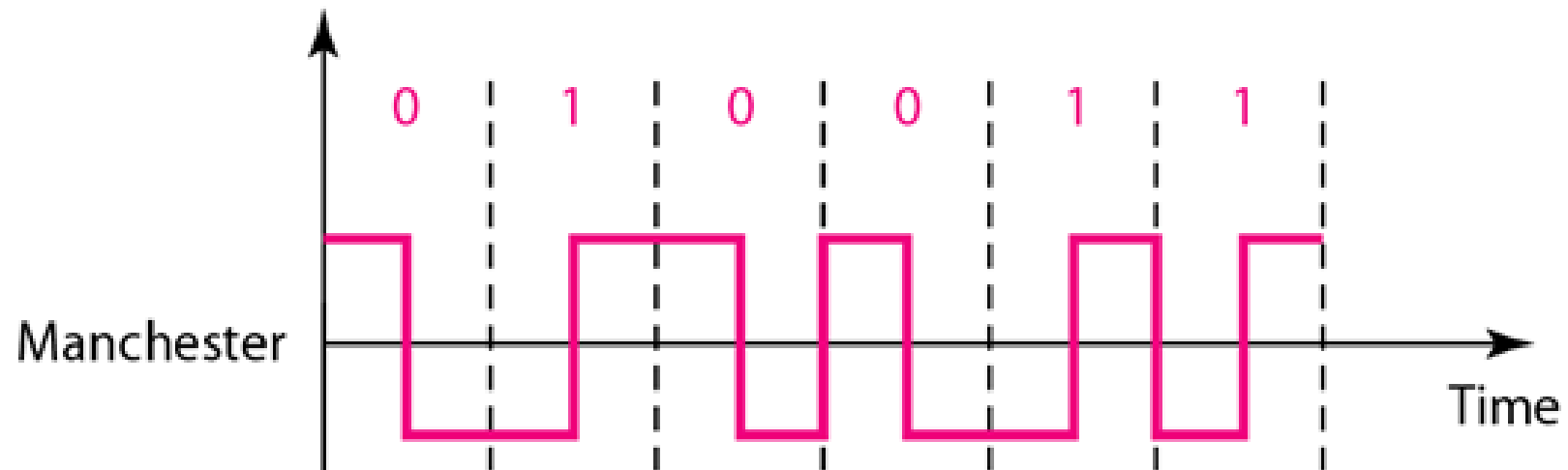
POLAR

- ❑ In case of **RZ**, the signal return to zero at $T/2$ duration of the signal for both bit 1 & 0 unlike unipolar. Similar to NRZ-L bit 1 = +V and bit 0 = -V.



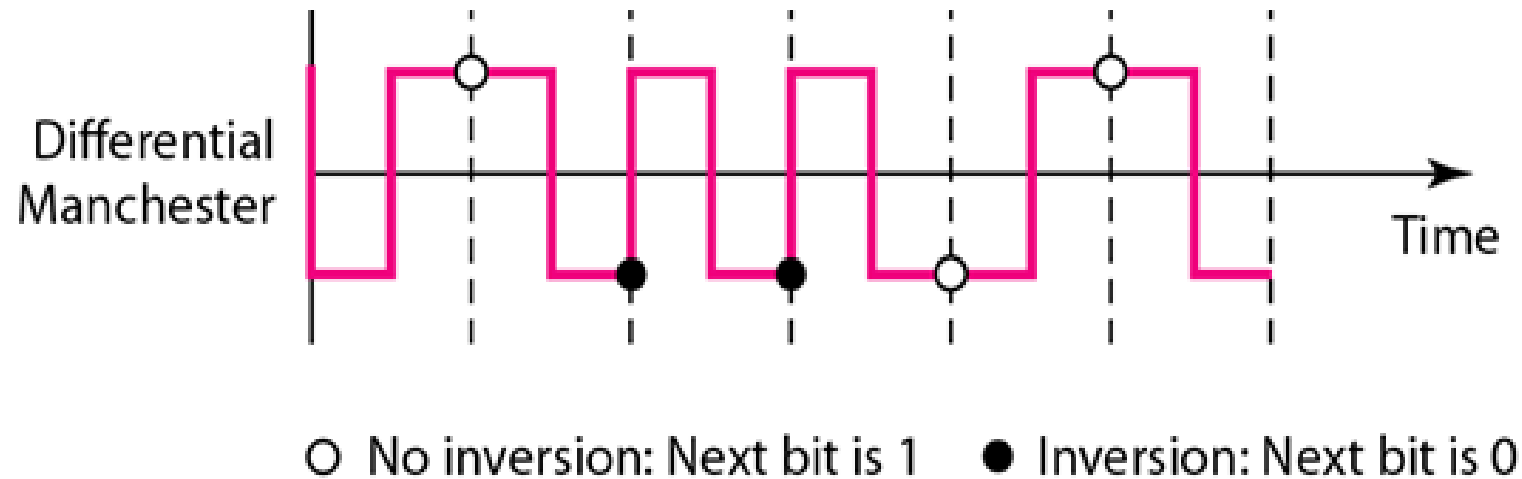
MANCHESTER

0 is  1 is 



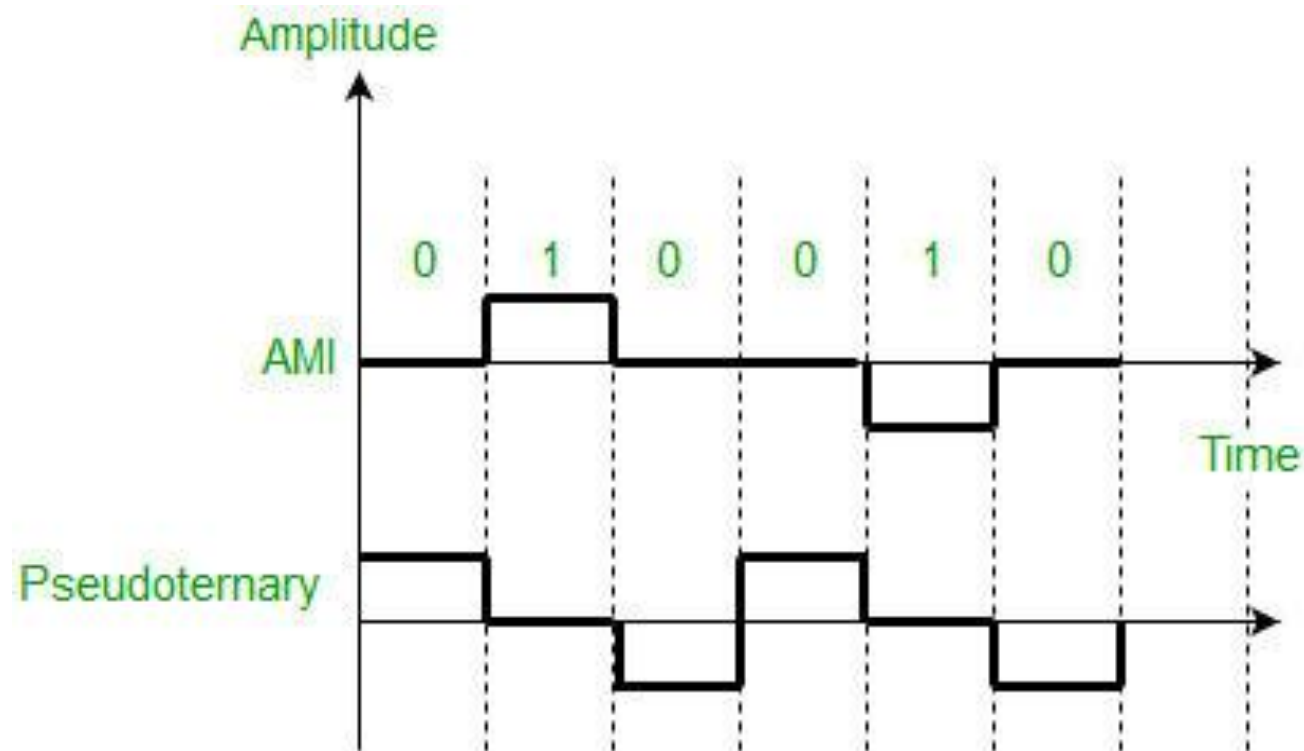
DIFFERENTIAL MANCHESTER

- ❑ In case of **D-Man**, the signal proceeds as it is when the next bit is 1 meaning there will be no inversion or transition and signal will continue from where the previous bit ended. But if the next bit is 0, then the signal continues from the positive side along the amplitude axis.



BIPOLAR

- ❑ **Alternate Mark Inversion (AMI):** Bit 0 represents zero voltage and bit 1 alternates between positive and negative voltage.
- ❑ **Pseudoternary:** Bit 1 represents zero voltage and bit 0 alternates between positive and negative voltage. This is opposite of AMI scheme.



PRACTICE PROBLEM

- ❖ Draw line coding (Man, D-Man, AMI, Pseudoternary, Polar NRZ-L & NRZ-I) for given data sequence: **001101**
- ❖ Sketch unipolar RZ and NRZ line coding for 11101011.

Check similar problems.



THANK YOU

